

Question number	Answer	Additional guidance	Mark
*7(a)	Indicative content: Graph: • HDL involved in removal of LDL from blood by liver (SE) • not just level of HDL that influences risk (D) • higher levels of LDL increase risk (D) • risk decreases with increase in HDL (D) • HDL results in uptake of cholesterol by liver (SE) • LDL forms the plaque (SE) • risk is a {combination / ratio} of HDL and LDL levels (SE) • the higher the HDL:LDL ratio the lower the risk (SE) Other risk factors: • the more cholesterol in the blood the more cholesterol to build up the atheroma (SE) • high blood pressure increases chance of damage to endothelial cell layer (SE) • smoking causes high blood pressure (SE) • smoking produces CO that binds to haemoglobin putting strain on heart (SE) • high salt in diet causes high blood pressure (SE) • obesity increases risk because of strain put on heart (SE) • age increases risk because affects heart muscle (SE) • genetic predisposition affects risk (SE) Extended explanation: • damage to the endothelial layer results in an inflammatory response • damage to the endothelial layer causes {build up of cholesterol / plaque} • plaque causes {more cholesterol to build up / blood clot to form • <u>coronary artery</u> becomes blocked • resulting in less oxygen reaching <u>heart {cells / tissues}</u> • heart attack results as <u>heart cells</u> cannot respire	Level 1: 1 mark = 1 {risk factor named / comment on graph} 2 marks = 2 {risk factor named / comment on graph} Or outline of how cholesterol causes CVD ie recall of the story Level 2: 3 marks = 2 simple explanations 4 marks = 3 simple explanations Level 3: 5 marks = as for 4 marks + 1 point from extended explanation 6 marks = as for 5 marks + 2 points from extended explanation	(6)

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7(b)(i)	An answer that includes three of the following points, one of which must be a similarity for full marks: Similarities • both contain ApoA-1(1) • both contain a phospholipid layer (1) Differences • altered HDL is larger (1) • altered HDL has more CE (1) • altered HDL has fewer (long-chain polyunsaturated) PC (1)	ACCEPT converse throughout DO NOT PIECE TOGETHER ACCEPT altered HDL has no {ApoM / S1P / RBP4/ CRABP1}	(3)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes the following points: • less reduction (in the number) of free radicals (1) • therefore {cell damage / damage to lining of blood vessels / oxidative stress} will not be reduced (1) • therefore formation of {plaque / atheroma} will not be reduced (1)	ACCEPT free radicals will not be {reduced / neutralised} IGNORE free radicals won't increase ACCEPT {cell damage / oxidative stress} will occur ACCEPT more plaques NB max 1 mark if not in context of reduced antioxidant properties	(3)